

Development of Interactive Cyber Threat Visualization Dashboard

Abstract

The increasing complexity and frequency of cyber threats have made traditional monitoring methods insufficient.

This project proposes an interactive dashboard that visualizes cybersecurity incidents using geospatial maps, trend analysis charts, and hierarchical vulnerability structures. The dashboard assists analysts in detecting attack patterns and making proactive security decisions.

Introduction

Organizations rely heavily on digital infrastructure, increasing exposure to cyber threats.

Visualization dashboards help simplify large volumes of security data into meaningful insights.

Problem Statement

Security analysts struggle to interpret large volumes of cybersecurity data from multiple sources.

Traditional tools mainly present raw logs and tabular reports, limiting efficient threat pattern identification.

Objectives

1. Collect and preprocess cybersecurity data.
2. Design interactive visualizations.
3. Implement geospatial mapping of threats.
4. Categorize vulnerabilities using MITRE ATT&CK framework.
5. Generate executive-level reports.

Methodology

The project follows four stages:

1. Data Acquisition – Collect CVE feeds and simulated attack logs.
2. Data Preprocessing – Clean and normalize datasets.
3. Visualization Development – Create charts and maps.
4. Dashboard Integration – Integrate components using Plotly Dash.

Tools and Technologies

Programming Language: Python

Framework: Plotly Dash

Data Processing: Pandas, NumPy

Database: MySQL / MongoDB

Threat Data Source: CVE feeds, Simulated Logs

Conclusion

The dashboard transforms complex cybersecurity data into visual insights, enabling faster threat detection and improved decision-making.